AADINATH R

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PROFESSIONAL SUMMARY

B.Tech CSE graduate specializing in AI & ML, with hands-on experience in Python, FastAPI, full-stack development, and LLM-powered applications. Skilled in building AI-integrated solutions, REST APIs, and database-driven systems. Seeking an entry-level software development or AI engineering role.

TECHNICAL SKILLS

Programming Languages: Python, C, SQL, TypeScript

Databases & Backend: MySQL, PostgreSQL, SQLAlchemy, FastAPI, REST APIs, Pydantic, Fast API

Web Technologies: HTML, CSS, JavaScript, React, Tailwind CSS

Tools & Technologies: Git, GitHub, VS Code, Docker, Claude, Argon2, RAG

EDUCATION

B.Tech – Computer Science and Engineering (AI & ML) **2022 – 2026**

St. Thomas College of Engineering and Technology, Chengannur, Kerala [**CGPA: 7.86**]

Higher Secondary Education (CBSE) **2021 – 2022**

St. Mary's Residential Public School, Tiruvalla [**72%**]

INTERNSHIPS

Front-End Intern - DataPMI Technologies LLP **June 2025**

- Assisted in collaborative product development for Prolex AI.
- Gained hands-on experience with HTML, CSS, and JavaScript, along with teamwork and problem-solving.

Software Development Intern - Zyncrypt Technologies **July 2024**

- Gained practical exposure to project tracking and project-related activities.

Machine Learning Intern - Intrainz Innovation Pvt Ltd **Feb 2024**

- Gained hands-on exposure to machine learning concepts, data preprocessing, and model development.

PROJECTS

AuditSum - Governed RBAC LLM Summarization Service GitHub Repository

Python, FastAPI, SQLAlchemy, PostgreSQL, Argon2, Docker, LLM

- Built an LLM summarization service using FastAPI and Groq, with source-backed citations.
- Role-based access control (RBAC) with endpoint-level authorization and session-based authentication
- Developed policy controls for file type and size validation, regex-based PII detection and redaction, an administrator-controlled model kill switch, and per-IP rate limiting.
- Engineered a tamper-evident, SHA-256 hash-chained audit log with PostgreSQL-enforced append-only constraints, concurrency-safe writes, and automated integrity checks.
- Implemented citation verification against source documents.
- Built a map-reduce summarization pipeline with document chunking, retry and repair logic, a daily token-budget circuit breaker, and deterministic mock-provider fallback.
- Developed auditor and administrator dashboards for audit inspection, chain verification, policy decisions, document and run management, and runtime controls and containerized using Docker.

ResumeIQ - Multi-Provider AI Resume Matching Engine GitHub Repository

Python, FastAPI, React, TypeScript, PostgreSQL, Pydantic, Ollama, Groq API, OpenAI API

- Developed a local-first AI-powered resume matching platform integrating Ollama, Groq, and OpenAI through a unified provider architecture for structured resume and job-description extraction.
- Engineered a deterministic weighted scoring engine with canonical skill normalization, requirement classification, and citation-based evidence verification to ensure explainable compatibility assessments.
- Implemented Pydantic schema validation, human-in-the-loop extraction review, and automated validation to identify unsupported matches and maintain data integrity.
- Designed an interactive React dashboard featuring category-wise compatibility scores, requirement-level evidence tracing, resume improvement simulation, and AI execution telemetry (latency, token usage, retries, and validation status).

PaperLens - Hallucination Resistant RAG Document Assistant GitHub Repository

Python, FastAPI, FAISS, Sentence Transformers, LLM, HTML, CSS, JavaScript

- Developed a Retrieval-Augmented Generation (RAG) assistant that answers questions from uploaded PDF and TXT documents using retrieved source context to reduce unsupported responses.
- Implemented context-grounded prompting with explicit fallback responses when information is absent from uploaded documents, along with instructions to ignore prompt injections embedded in document content.
- Built a semantic search pipeline using Sentence Transformers (all-MiniLM-L6-v2) and FAISS IndexFlatIP to retrieve the top four relevant text chunks using cosine similarity.
- Implemented document processing with PyMuPDF, 600-word chunking, 100-word overlap, and page-level metadata to support source traceability and citations.
- Integrated LLM functionality with FastAPI REST APIs and developed a responsive HTML/CSS/JavaScript interface supporting document uploads, question answering, source citations, retrieved-chunk inspection, and session reset.

EXTRACURRICULAR & LEADERSHIP

- **Flagship Coordinator, KTU CARE NSS Unit No. 553:** Led healthcare awareness campaigns and digital literacy outreach programs for underprivileged communities across Chengannur.
- **Class Representative (2023–2024):** Coordinated communication between students and faculty regarding academic activities at St. Thomas College of Engineering and Technology.

LANGUAGES

English (Professional) | Malayalam (Native) | Hindi (Conversational)